

L10a: Monte Carlo Pricing and Sensitivity Analysis of American Options Premium

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Objectives for Today

Previously, we priced American options on a CRR tree. Today, we:

Today's concepts

- Estimate and decompose the premium of American options: Longstaff–Schwartz Monte Carlo, then intrinsic and extrinsic value.
- Compute the Greeks for American options: finite differences on a binomial tree.
- Interpret Delta and compute the probabilities of exercise and profit: Delta as a share equivalent, compared with the POE, and both probabilities from simulated share prices.

What You Need to Know

Core results

- Premium $=$ intrinsic value $+$ extrinsic value.
- What each Greek measures. Per day: Theta$/365$. Per percentage point: Vega$/100$ and Rho$/100$.
- The probabilities of exercise (POE) and profit (POP) come from simulated share prices at expiration. For a short-dated call (no dividends), Delta is close to the POE under the risk-neutral $\mathbb{Q}$.

For understanding (you won't reproduce these)

- The Longstaff–Schwartz regression, the Taylor expansion, and the Delta–POE proposition.

Lectures and Examples

Lecture:
CHEME-5660-L10a-Lecture-American-Derivatives-Greeks-Fall-2026.ipynb

Example 1 (review on your own):
CHEME-5660-L10a-Example-LSM-Premium-Fall-2026.ipynb

Example 2:
CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

Example 3:
CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

Review on your own:
CHEME-5660-L10a-Example-ExtrinsicValue-Fall-2026.ipynb

Company Profile: Susquehanna

Susquehanna (SIG): a privately held derivatives trading firm, started in 1987 by options floor traders in Philadelphia.

- Options market making: it takes the other side of its customers' contracts. In October 2026, it was the Designated Primary Market Maker for 1,359 of 5,370 Cboe option classes, the most of any firm.
- Managing the Greeks: it nets Delta, Vega, Theta, and Rho across its book.
- Training: options theory and poker for deciding under uncertainty.

Connection: when a customer buys a call, SIG hedges with $100\,\Delta_c$ shares and rebalances as Gamma moves Δ_c.

Monte Carlo Pricing of American Options

We simulate share price paths under $\mathbb{Q}$ and, like the lattice, work backward from expiration. At each date, a regression on the share price estimates the value of holding.

Initialize: given the share price S_0, strike price K, payoff $V(K, S)$ of the call or put, time to expiration T (years), volatility σ, risk-free rate g_y, number of exercise dates h after today, and number of paths M. Compute $\Delta t = T/h$ and the one-step discount factor $\mathcal{D}_{t+1,t}^{-1}(g_y) = e^{-g_y \Delta t}$.

Reference: Longstaff, F. A., and Schwartz, E. S. (2001). Valuing American options by simulation: a simple least-squares approach. Review of Financial Studies 14(1): 113–147.

Pseudocode: Longstaff–Schwartz

1. Simulate M paths under $\mathbb{Q}$ from S_0 on the dates $t = 0, 1, \ldots, h$; $S_i(t)$ is the share price on path i at date t.
2. At expiration, each path's cash flow is its payoff: $Y_i(h) \leftarrow V(K, S_i(h))$ for $i = 1, 2, \ldots, M$, where $Y_i(t)$ is path i's cash flow valued at date t.
3. For each date $t = h-1, h-2, \ldots, 1$ **do**:
   a. Discount one date: $Y_i(t) \leftarrow \mathcal{D}_{t+1,t}^{-1}(g_y) \cdot Y_i(t+1)$.
   b. Regress: on the in-the-money paths ($V(K, S_i(t)) > 0$), fit $\hat{H}_t(S) = \beta_0 + \beta_1 S + \beta_2 S^2$ to the pairs $(S_i(t), Y_i(t))$ by least squares.
   c. Decide: on each in-the-money path, if $V(K, S_i(t)) > \hat{H}_t(S_i(t))$, **then** exercise: $Y_i(t) \leftarrow V(K, S_i(t))$.
4. Discount to today and average: $\bar{Y} \leftarrow \mathcal{D}_{1,0}^{-1}(g_y) \cdot \frac{1}{M} \sum_{i=1}^{M} Y_i(1)$.
5. The premium is $\mathcal{P}^{\text{AM}} \leftarrow \max\{V(K, S_0),\ \bar{Y}\}$.

The Early-Exercise Premium by Simulation

We price an American put by applying two rules to the same simulated paths:

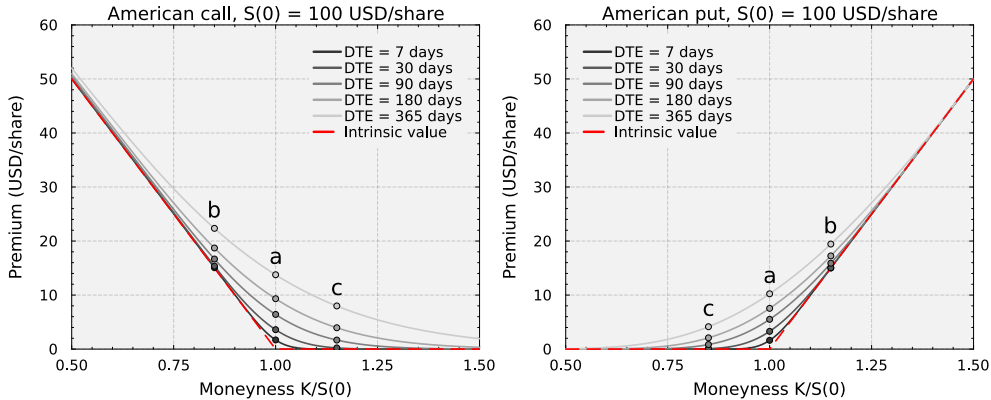
Rule at each date	Estimates
Longstaff–Schwartz: exercise when $V(K, S_i(t)) > \hat{H}_t(S_i(t))$	$\mathcal{P}^{\text{AM}}$
Hold on every path (European)	$\mathcal{P}^{\text{EU}}$

$$\text{early-exercise premium} \approx \mathcal{P}^{\text{AM}} - \mathcal{P}^{\text{EU}}$$

Example 1 (review on your own): CHEME-5660-L10a-Example-LSM-Premium-Fall-2026.ipynb

Extrinsic versus Intrinsic Value

American call (left) and put (right) premiums against the moneyness $K/S(0)$ for several days to expiration (DTE). Point a is at the money, b in the money, c out of the money.



Intrinsic and Extrinsic Value

The premium of an American option has two parts:

- Intrinsic value: what exercising today pays (red dashed line), $\max(0, S(t) - K)$ for a call and $\max(0, K - S(t))$ for a put, with share price $S(t)$ and strike price K.
- Extrinsic value (time value): the rest of the premium, the gap above the red dashed line.

$$E_V(t) = \text{Option Price} - \text{Intrinsic Value}$$

Extrinsic Value

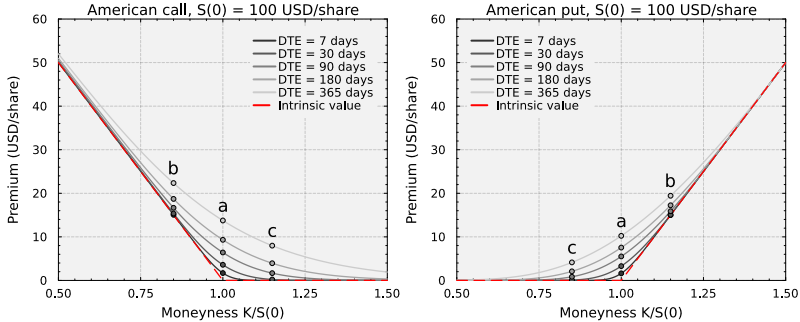
With call and put premiums $\mathcal{P}_c$ and $\mathcal{P}_p$, the extrinsic value is:

$$E_V(t) = \begin{cases} \mathcal{P}_c(K, S(t)) - \max(0, S(t) - K) & \text{(for Call Options)} \\ \mathcal{P}_p(K, S(t)) - \max(0, K - S(t)) & \text{(for Put Options)} \end{cases}$$

- At or out of the money (points a, c): no intrinsic value, so the whole premium is extrinsic, largest at a.
- Time decay: extrinsic value shrinks as expiration nears.
- Why pay for “worthless”? The share price can still move favorably before expiration.

Would You Ever Exercise Early?

- A call: no, not without dividends. Exercising buys the shares at K today. Holding, we pay K later (earning interest on it) and skip buying if the share price falls below K.
- A put: yes, far in the money. Exercising sells the shares at K today, and that cash earns interest.



What If the Share Pays a Dividend?

The dividend goes to those who own the shares before the ex-dividend date, not to call holders.

If the dividend is larger than the extrinsic value we give up, which is small far in the money, we exercise the trading day before the ex-dividend date.

Review on your own: CHEME-5660-L10a-Example-ExtrinsicValue-Fall-2026.ipynb

The Greeks

The Greeks measure how the premium changes with market conditions:

- Delta Δ: premium change per `+1 USD/share` in the share price.
- Theta Θ: premium change per `1-day decrease` in the contract's lifetime (time decay), quoted as $\Theta_\star/365$.
- Vega ν: premium change per `+1 point` in volatility, quoted as $\nu_\star/100$.
- Gamma Γ: change in Δ per `+1 USD/share` in the share price.
- Rho ρ: premium change per `+1 point` in the risk-free rate, quoted as $\rho_\star/100$; minor except for long-dated options.

Trading platforms compute these using the implied volatility under $\mathbb{Q}$.

A Local Taylor Expansion of the Premium

Expanding the premium $\mathcal{P}_\star(S, \sigma, t, g_y)$ of a contract of type $\star \in \{c, p\}$ at calendar time t to second order in S and first order in the other inputs gives:

$$\begin{aligned}\Delta\mathcal{P}_\star &\approx \Delta_\star\,\Delta S + \nu_\star\,\Delta\sigma + \Theta_\star\,\Delta t + \rho_\star\,\Delta g_y + \tfrac{1}{2}\,\Gamma_\star\,(\Delta S)^2 \\ &= \underbrace{(\Delta_\star + \tfrac{1}{2}\,\Gamma_\star\,\Delta S)\Delta S}_{\text{share price}} + \underbrace{\nu_\star\,\Delta\sigma}_{\text{volatility}} + \underbrace{\Theta_\star\,\Delta t}_{\text{time}} + \underbrace{\rho_\star\,\Delta g_y}_{\text{rate}}\end{aligned}$$

$$\Delta_\star = \frac{\partial \mathcal{P}_\star}{\partial S} \qquad \nu_\star = \frac{\partial \mathcal{P}_\star}{\partial \sigma} \qquad \Theta_\star = \frac{\partial \mathcal{P}_\star}{\partial t} \qquad \rho_\star = \frac{\partial \mathcal{P}_\star}{\partial g_y} \qquad \Gamma_\star = \frac{\partial^2 \mathcal{P}_\star}{\partial S^2}$$

One day changes the premium by $\Theta_\star/365$; a one-point rise in the volatility or the rate changes it by $\nu_\star/100$ or $\rho_\star/100$.

Delta and Gamma

Delta $\Delta_\star$ and Gamma $\Gamma_\star$ are the slope and curvature of the premium against the share price.

Example: a long call with $\Delta_c = 0.50$ and $\Gamma_c = 0.02$ per USD/share:

$$\Delta S = +5: \quad \Delta \mathcal{P}_c \approx (0.50 + \tfrac{1}{2}(0.02)(5)) \times 5 = 0.55 \times 5 = 2.75 \text{ USD/share}$$
$$\Delta S = -5: \quad \Delta \mathcal{P}_c \approx (0.50 - 0.05) \times (-5) = -2.25 \text{ USD/share}$$

Gamma adds $\frac{1}{2}\Gamma_c(\Delta S)^2 = 0.25$ USD/share to the Delta-only ± 2.50: the call gains more going up and loses less going down.

Computing the Greeks of American Options

The tree gives no formula to differentiate, so we reprice it with one input moved up and down by ϵ, a central finite difference:

$$\frac{\partial \mathcal{P}_\star}{\partial x} \approx \frac{\mathcal{P}_\star(x+\epsilon) - \mathcal{P}_\star(x-\epsilon)}{2\epsilon} \qquad x \in \{S, \sigma, t, g_y\}$$

$$\Gamma_\star \approx \frac{\mathcal{P}_\star(S+\epsilon) - 2\,\mathcal{P}_\star(S) + \mathcal{P}_\star(S-\epsilon)}{\epsilon^2}$$

With enough tree steps, the CRR tree estimates Delta well, but its Gamma can be zero or erratic for a small step ϵ, so we use the Leisen–Reimer tree, a modified CRR tree.

Example 2: CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

The Secret Lives of Delta

Besides the premium change per `+1 USD/share`, Δ has two other meanings:

- Share equivalent: n contracts ($n < 0$ if sold) gain or lose as much as $100 \cdot n \cdot \Delta$ shares for small moves. Holding $-100 \cdot n \cdot \Delta$ shares makes the position delta-neutral, so Delta is the hedge ratio.
- Probability of exercise: Delta estimates the probability of expiring in the money; a call with $\Delta = 0.7$ suggests about 70% under $\mathbb{Q}$.

Probability of Exercise

Probability of exercise (POE): the probability that the contract finishes in the money at expiration $t = T$:

$$\text{POE} = \begin{cases} \mathbb{P}\left(S_T > K + 0.01\right) & \text{(for Call Options)} \\ \mathbb{P}\left(S_T < K - 0.01\right) & \text{(for Put Options)} \end{cases}$$

where S_T is the share price at expiration and K is the strike price.

Why 0.01 USD/share? OCC automatically exercises equity options in the money by 0.01 USD/share or more.

Why Is Delta Close to the POE?

Quoted Deltas come from pricing models under $\mathbb{Q}$, so we compare Delta with the POE under $\mathbb{Q}$. **Proposition:** for a call (no dividends) on an h-step CRR tree under $\mathbb{Q}$, both sum over the same in-the-money nodes, ITM $= \{k : S_0 u^k d^{h-k} > K\}$ with k up moves:

$$\Delta_c = \sum_{k \in \text{ITM}} \binom{h}{k} (q^*)^k (1-q^*)^{h-k} \qquad \text{quoted Delta: up probability } q^*$$

$$\mathbb{Q}(S_T > K) = \sum_{k \in \text{ITM}} \binom{h}{k} q^k (1-q)^{h-k} \qquad \text{POE under } \mathbb{Q}\text{: up probability } q$$

Here u, d, q, and Δt are the tree's up and down factors, up probability, and step, and $q^* = q\,u\,e^{-g_y \Delta t}$ is q times u over the risk-free growth $e^{g_y \Delta t}$ in one step. Delta is larger than the POE under $\mathbb{Q}$: $u > e^{g_y \Delta t}$, so $q^* > q$.

Derivation: advanced/delta_poe/CHEME-5660-L10a-Derivation-Delta-POE-Fall-2026.ipynb

When Is Delta Close to the POE?

- Volatility and time to expiration: the gap grows with both, so Delta is close to the POE under $\mathbb{Q}$ for short-dated contracts.
- Our forecast and American puts: our forecast POE under $\mathbb{P}$ puts the real-world up probability p in place of q. Nothing ties p to q^*, so it can sit above or below Delta: Delta follows $\mathbb{Q}$, not our belief. Early exercise adds to an American put's $|\Delta_p|$, so the match is approximate.

Probability of Profit

Probability of profit (POP): the probability that the profit at expiration $\hat{P}_{\mathcal{C}}$ is positive:

$$\boxed{\text{POP} = \mathbb{P}\left(\hat{P}_{\mathcal{C}} > 0\right)}$$

For a long call, the profit is the payoff minus the premium's future value:

$$\begin{aligned}
\hat{P}_{\mathcal{C}} > 0 &\iff (S_T - K)^+ - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c > 0 && \text{profit of a long call} \\
&\iff (S_T - K)^+ > \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c && \text{add the premium's future value} \\
&\iff S_T - K > \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c && \text{a positive payoff needs } S_T > K \\
&\iff S_T > \underbrace{K + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c}_{= S^\ddagger} && \text{the breakeven share price}
\end{aligned}$$

Probability of Profit and the Breakeven

For a put, $S_T < S^\ddagger$ with $S^\ddagger = K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$. The seller profits exactly when the buyer does not:

$$\text{POP} = \begin{cases} \mathbb{P}\left(S_T > S^\ddagger\right) & \text{(bought a call or sold a put)} \\ \mathbb{P}\left(S_T < S^\ddagger\right) & \text{(sold a call or bought a put)} \end{cases}$$

The POP is the POE with the breakeven in place of the strike, so the buyer's POP is smaller than the POE. We compute both under our belief $\mathbb{P}$: GBM with our estimates of μ and σ.

Computing POE and POP by Simulation

Initialize: given S_0, K, the premium $\mathcal{P}$ of the call or put, T (years), the drift μ and volatility σ under $\mathbb{P}$, g_y, and the number of samples M. The breakeven is $S^{\ddagger} = K + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}$ for a call or $S^{\ddagger} = K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}$ for a put, with $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$.

1. For each sample $i = 1, 2, \ldots, M$ **do**:
   a. Draw $Z_i \sim \mathcal{N}(0,1)$.
   b. Compute $S_i(T) \leftarrow S_0 \exp\left[(\mu - \sigma^2/2)\,T + \sigma\sqrt{T}\,Z_i\right]$.
2. Compute the fraction of samples at or below each share price x: $\widehat{F}(x) \leftarrow \frac{1}{M} \cdot \#\{i : S_i(T) \le x\}$.

Pseudocode: POE and POP

3. If the contract is a call, **then** POE $\leftarrow 1 - \hat{F}(K + 0.01)$. Otherwise, POE $\leftarrow \hat{F}(K - 0.01)$.
4. Compute the probability of profit:
   - If we bought a call or sold a put, **then** POP $\leftarrow 1 - \hat{F}(S^{\ddagger})$.
   - If we sold a call or bought a put, **then** POP $\leftarrow \hat{F}(S^{\ddagger})$.

With $\mu = g_y$ and σ set to the implied volatility, the same steps give the probabilities under $\mathbb{Q}$. These are the probabilities inside the market price, not our forecast.

Example 3: CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

Optional Advanced Material

Leisen–Reimer tree: advanced/leisen_reimer/
CHEME-5660-L10a-Advanced-LeisenReimer-Greeks-Fall-2026.ipynb

When can repricing the CRR tree give a Gamma of zero, and how does a tree built around the strike avoid it? We derive the Leisen–Reimer tree and compute the Greeks on it.

Summary

Key takeaways

- Estimating and decomposing the American premium: we regressed the value of holding on the share price along simulated paths. Extrinsic value is largest at the money; without dividends, only a deep in-the-money put is worth exercising early.
- The Greeks quantify option price sensitivity: we estimated them by finite differences on a binomial tree, as coefficients of a local Taylor expansion of the premium.
- Delta serves three purposes: premium change per USD/share, share equivalent, and an estimate of a call's POE. The POP is the POE with the breakeven in place of the strike.